Tutorial 2

Ex. 8.2

Q23 $\displaystyle\int \underbrace{e^{2x}}_{\text{int}} \underbrace{\cos 3x}_{\text{diff}}\, dx$

$$= \frac{1}{2}e^{2x}\cos 3x$$

$$- (-3\sin 3x)\frac{1}{4}e^{2x}$$

$$-\frac{9}{4}\int e^{2x}\cos 3x\, dx$$

Tabular Integration

	diff	int
+	$\cos 3x$	e^{2x}
−	$-3\sin 3x$	$\frac{1}{2}e^{2x}$
+	$-9\cos 3x$	$\frac{1}{4}e^{2x}$

$$\boxed{+\int (-9\cos 3x)\left(\frac{1}{4}e^{2x}\right) dx}$$

Let $I := \displaystyle\int e^{2x}\cos 3x\, dx.$

Then $I = \frac{1}{2}e^{2x}\cos 3x + \frac{3}{4}e^{2x}\sin 3x - \frac{9}{4}I$

$I = \cdots \qquad + C$

Q28 $\int 1 \cdot \ln(x+x^2)\,dx$

$$= x\ln(x+x^2) - \int \frac{1+2x}{x+x^2}\cdot x\,dx$$

$$\frac{1+2x}{x+x^2}\cdot x$$

$$= \frac{x+2x^2}{x+x^2} \rightarrow \frac{2(x+x^2)}{x+x^2} + \frac{\text{Extra}}{x+x^2}$$

$$= \frac{x+2x^2 + x - x}{x+x^2}$$

$$= \underbrace{\frac{2x+2x^2}{x+x^2}}_{2} - \underbrace{\frac{x}{x+x^2}}_{\frac{1}{1+x}}$$

$u = x + x^2 \rightarrow 2x \rightarrow$ CTS

$u = 1+2x$ $\qquad (x+a)^2 + b = u$

$$\int \ln(x+x^2)\,dx$$

$$= \int \ln u \cdot \frac{1}{\boxed{1+2x}}\,du$$

$$\frac{1}{1+2x}$$

$$\frac{d}{dx}\ln(x+x^2) = \frac{1+2x}{x+x^2}$$

$$\int 2\,dx = x$$

$$\int \frac{1+2x}{x+x^2}\cdot x\,dx = 2x - \ln|1+x| + C$$

Q54 $\displaystyle\int \frac{xe^x}{(x+1)^2}\,dx$

$u = x+1 \Rightarrow x = u-1$

$du = dx$

$$= \int \frac{(u-1)e^{u-1}}{u^2}\,du$$

$$= \int \frac{ue^{u-1}}{u^2}\,du - \int \frac{e^{u-1}}{u^2}\,du$$

$$= \int \underbrace{u^{-1}}_{\text{diff}}\underbrace{e^{u-1}}_{\text{int}}\,du - \int \frac{e^{u-1}}{u^2}\,du$$

$\frac{d}{du}(u^{-1}) = -u^{-2}$

$\int e^{u-1}\,du = e^{u-1}$

$$= u^{-1}e^{u-1} - \int -u^{-2}e^{u-1}\,du - \int \frac{e^{u-1}}{u^2}\,du$$

$$= u^{-1}e^{u-1} + \int u^{-2}e^{u-1}\,du - \int u^{-2}e^{u-1}\,du$$

$$= u^{-1}e^{u-1} + C = (x+1)^{-1}e^{x} + C$$

Try Q53

Ex 5.3

$u = \sin t \Rightarrow du = \cos t \, dt$

Q16 $\int 7\cos^7 t \, dt = \int 7\cos^6 t \cdot \cos t \, dt$

$\downarrow$

$(\cos^2 t)^3$

$\hookrightarrow 1 - \sin^2 t$

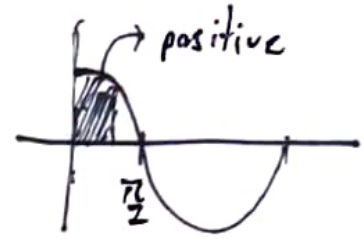
$1 - u^2$

Q18 $\int 8\cos^4 2\pi x \, dx = \int 8 \left(\frac{1 + \cos 4\pi x}{2} \right)^2 dx$

$\cos^2 \theta = \frac{1 + \cos 2\theta}{2}$

Q28

$$\int_0^{\pi/6} \sqrt{1+\sin x}\, dx \cdot \frac{\sqrt{1-\sin x}}{\sqrt{1-\sin x}}$$



$$= \int_0^{\pi/6} \frac{\sqrt{1-\sin^2 x}}{\sqrt{1-\sin x}} \longrightarrow 1-\sin^2 x = \cos^2 x$$

$$\Rightarrow \sqrt{1-\sin^2 x} = \sqrt{\cos^2 x} = |\cos x|$$

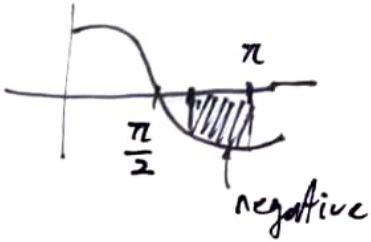
$$\therefore \sqrt{1-\sin^2 x} = \cos x$$

$$\because 0 \leq x \leq \pi/6$$

$$= \int_0^{\pi/6} \frac{\cos x}{\sqrt{1-\sin x}}\, dx$$

$$u = 1 - \sin x \Rightarrow du = -\cos x\, dx$$

Q2#9 $\displaystyle\int_{5\pi/6}^{\pi} \frac{\cos^4 x}{\sqrt{1-\sin x}}\,dx \cdot \frac{\sqrt{1+\sin x}}{\sqrt{1+\sin x}}$



$$= \int_{5\pi/6}^{\pi} \frac{\cos^4 x\sqrt{1+\sin x}}{\sqrt{1-\sin^2 x}}\,dx \longrightarrow \sqrt{1-\sin^2 x} = |\cos x|$$

$\cos x$ $\quad$ $\boxed{-\cos x}$

$$= \int_{5\pi/6}^{\pi} \frac{\cos^4 x\sqrt{1+\sin x}}{-\cos x}\,dx$$

$$= \int_{5\pi/6}^{\pi} -\cos^3 x\sqrt{1+\sin x}\,dx$$

$u = 1+\sin x$

$du = \cos x\,dx$

$$= \int_{\square}^{\square} -\cos^2 x\sqrt{u}\,du$$

$1-\sin^2 x = 1-(u-1)^2$

Q60 $\int \cos^2 2\theta \sin\theta \, d\theta$

$$\cos^2 x = \frac{1+\cos 2x}{2}$$

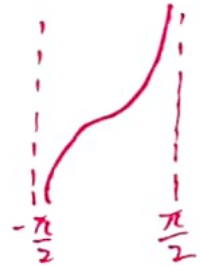
$$\left(\frac{1+\cos 4\theta}{2}\right)\sin\theta$$

$$\frac{1}{2}\sin\theta \qquad \frac{1}{2}\underbrace{\cos 4\theta \sin\theta}_{\text{Product to Sum Formula}}$$

$$\sin A \cos B = \frac{1}{2}\sin(A-B) + \frac{1}{2}\sin(A+B)$$

Ex. 8.4

$1 + x^2 \rightarrow x = \tan\theta \qquad -\frac{\pi}{2} < \theta < \frac{\pi}{2}$



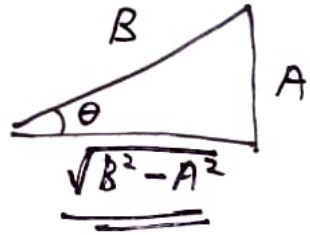
$\sqrt{1-x^2} \rightarrow x = \sin\theta \qquad -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$

$\sqrt{x^2-1} \rightarrow x = \sec\theta \qquad \boxed{0 \leq \theta < \frac{\pi}{2}}$

$\hookrightarrow \sqrt{\tan^2\theta} = |\tan\theta| = \tan\theta$

$\sin\theta = \frac{A}{B}$

$\cos\theta = ?$



Q48

$$\int \frac{\sqrt{x-2}}{\sqrt{x-1}}\,dx \qquad u = x-1$$

$$= \int \frac{\sqrt{u-1}}{\sqrt{u}}\,du \qquad \underline{t = \sqrt{u}} \Rightarrow u = t^2$$

$$= \square \int \sqrt{u-1}\,dt$$

$$= \square \int \sqrt{t^2-1}\,dt$$

$$\Downarrow \quad t = \sec\theta \;,\; 0 \leq \theta < \frac{\pi}{2}$$